College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Final Examination of Semester 2

Subject: 392132 Physics II

Date: 17 May 2018

Year: 2017

Section: 15-18

Time: 10.00 - 12.00

Name:______________________ID:______________Class:________

Questions	Points
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
12	
13	
total	

Instructions:

1. The examination has 7 pages (including this page) and a total score of 60 points.
2. Write all your solutions and answers on this examination sheet.
3. This is a closed book examination.
4. You are not allowed to leave the exam room during the first 1 hour after the beginning of the exam.
5. You are not allowed to open the exam papers or start to answer before the proctor's permission.
6. You are not allowed to use the restroom during the exam except in case of an emergency.
7. No documents are allowed to be taken out of the examination room.
8. Calculators are **NOT** allowed in the examination.
9. Electronic communication devices are **NOT** allowed in the examination room.
10. If not specified otherwise, used these values: $g = 10\ m/s^2, \pi = 3.14, \pi^2 = 10$.

Cheating in the exam is considered an extremely serious offence which will result in expulsion from the University.

1.1 What are the magnitude and direction of the momentum of a garbage truck that is $1.2\times10^{4}\ kg$ and $10.0\ {}^{m}/_{s}$ (2 points)

1.2. A car mass 100 kg moving 15 m/s in west direction. What is the magnitude and direction of the momentum for this car? (2 Points)

2. Block A mass 5 kg moving 10 m/s in left hand and block B mass 20 kg moving 4 m/s in right hand. After block B collide with block A, Block A moving 2 m/s in right hand.

(a) What is velocity of block B after collide? (3 Points)

(b) What the total energy before collide? (2 Points)

(c) What the total energy after collide? (2 Points)

(d) This event, Is elastic collision or inelastic collision and because? (1 Point)

3. A tennis ball traveling at 10.0 m/s is returned by Venus Williams. It leaves her racket with a speed of 40.0 m/s in the opposite direction from which it came.

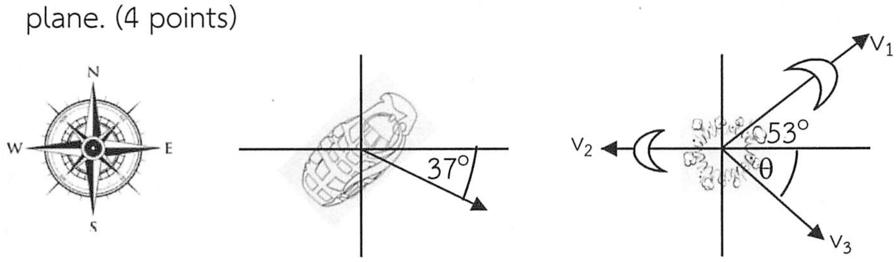
   a) What is the change in momentum of the tennis ball? (3 points)

   b) If the 0.050-kg the tennis ball is in contact with the racket for 0.020 s, with what average force does Venus hit? (3 points)

4. A 20-kg bomb moving at 5 m/s in the direction 37° Southeast explodes into three pieces. A 2-kg piece moves off at 10 m/s at 53° Northeast while a 5-kg piece moves West at 5 m/s.

   (a) What is the momentum before exploded?(3 points)

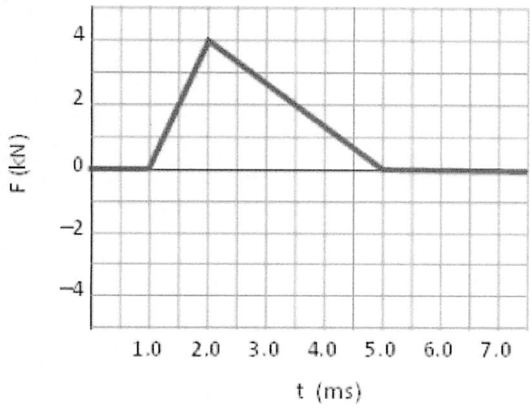
   (b) What is the velocity of the third fragment? Assume all motion occurs in a horizontal plane. (4 points)



5. Estimate the force and time of impact of the ball with the wall as shown in figure in below.

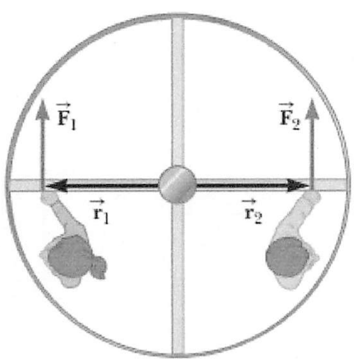
(a) What is the change of momentum of the ball? (3 points)

(b) What is the average force exerted on the ball and the wall?(2 points)



6. Why does a solid sphere have a smaller moment of inertia than a hollow sphere? If two of these spheres are the same mass and radius, and about axis pivot passing though their axis of symmetry? (2 Points)

7.

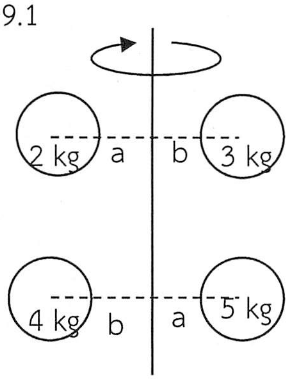


Two disgruntled business people are trying to use a revolving door. The woman on the left exerts a force of 500 N perpendicular to the door and 1.50 m from the hub's center, while the man on the right exerts a force of 15.0 x 10^2 N perpendicular to the door and 0.50 m from the hub's center. Find the net torque (both magnitude and direction) on the revolving door.(4 Points)

8. A ball with an initial velocity of 8 m/s rolls up a hill without slipping. Treating the ball as a spherical shell, calculate the vertical height it reaches (3 points)

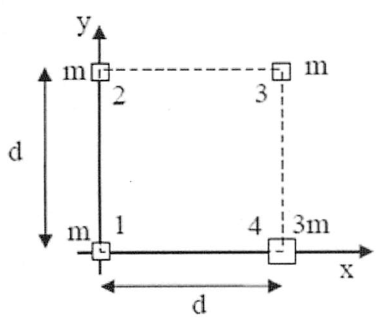
Assume : $I_{ball} = \frac{2}{3}mr^2$

9.1

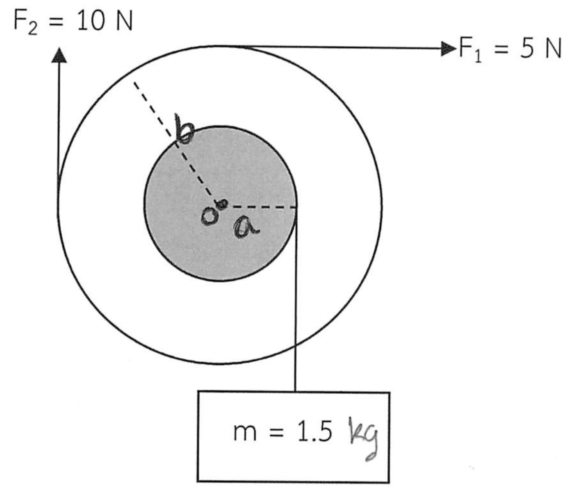


From the figure, What is moment of inertia of this system? (a = 2 m and b = 1 m)(3 points)

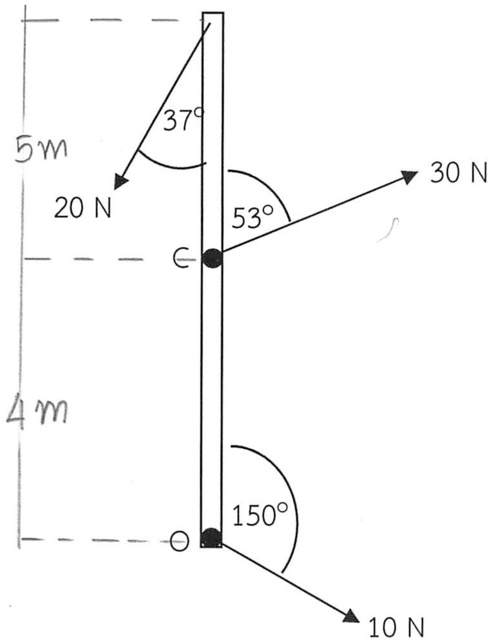
9.2 Where is moment of inertia of this 4 mass system rotate with y-axis ? The masses, labeled 1, 2, 3, 4, form a square of edge length d. The four masses are m, m, m, and 3m.(3 points)



10. Find the net torque on the wheel about the axle through point O perpendicular to the page, talking a = 20 m and b = 40 cm. (3 points)



11. Calculate the net torque (magnitude and direction) on about an axis through point O and an axis through point C perpendicular to the page. (Give : tan 53°= 4/5) (3 points)



12. A large grinding wheel in the shape of a solid cylinder with the of radius 0.2 m is free to rotate on a frictionless, vertical axle. A constant tangential force of 250 N applied to its edge causes the wheel to have an angular acceleration of 1.0 rad/s^2.

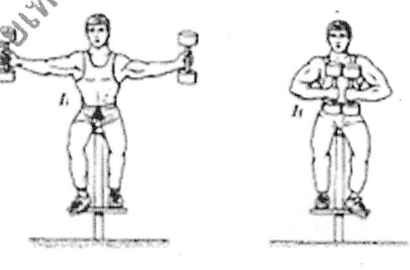
(a) What is the moment of inertia of the wheel?(3 points)

(b) What is the mass of the wheel? ($I = 0.5mR^2$)(2 points)

13. A student sits on a pivoted stool while holding a pair of weights. The stool is free to rotate about a vertical axis with negligible friction. The moment of inertia of student, weights, and stool is 2.00 kg.m^2. The student is set in rotation with arms outstretched, making one complete turn every 2 minutes, arms outstretched.

(a) What is the initial angular speed of the system?(2 points)

(b) As he rotates, he pulls the weights inwards so that the new moment of inertia of the system (student, objects and stool) becomes 1.25 kg.m^2. What is the new angular speed of the system?(2 points)



Instructor team

Asst.Prof.Dr. Prakasit CHANGSUPHAN

Asst.Prof.Dr. Dusit NGAMRUNGROJ

Dr. Sethavut DUANGCHAN